Nonexistence of integer solutions to $6 + x^3 - y^2 + y^2z^2 = 0$

Theorem

There are no integers x, y, z with z even, y odd, and $x \equiv 3 \pmod 4$ satisfying

$$6 + x^3 - y^2 + y^2z^2 = 0.$$

Proof

Rewrite the equation as

$$x^3 + 6 = -y^2(z^2 - 1). \tag{1}$$

We split into four exhaustive cases.

Case 1: $y = 0$. If $y = 0$ then (1) reduces to $x^3 = -6$. No integer cube equals -6, because $(-2)^3 = -8$ and $(-1)^3 = -1$. Hence there is no solution with $y = 0$.

Case 2: $z = 0$, $y \neq 0$. If $z = 0$ then (1) becomes

$$y^2 = x^3 + 6. \tag{2}$$

This is Mordell's equation with parameter 6. It is a classical result that this elliptic curve has no integer points. In particular, the curve with LMFDB label `24.a3` has rank 0 over $\mathbb{Q}$ and no rational points besides the point at infinity. Therefore there is no integer solution with $z = 0$ and $y \neq 0$.

Case 3: $z = \pm 1$, $y \neq 0$. If $z = \pm 1$ then $z^2 - 1 = 0$ and (1) reduces again to $x^3 = -6$. As in Case 1, this is impossible.

Case 4: $|z| \geq 2$, $y \neq 0$. Since z is even, $z^2 - 1$ is an odd integer and $z^2 - 1 \geq 3$. Because $y \neq 0$, the right-hand side of (1) is at most -3. Thus $6 + x^3 \leq -3$, which forces $x^3 \leq -9$ and therefore $x \leq -3$. Because $x \equiv 3 \pmod 4$, in fact $x \leq -5$. Set $X := -x \geq 5$, so $X \equiv 1 \pmod 4$.

Rewrite (1) as

$$X^3 + 6 = y^2(z^2 - 1). \tag{3}$$

The right-hand side is odd, so X is odd.

Since z is even, both $z - 1$ and $z + 1$ are odd and coprime. One of them is congruent to 1 modulo 4 and the other is congruent to 3 modulo 4. Because y is odd, $y^2 \equiv 1 \pmod 4$. Hence

$$\{(z-1)y^2,\ (z+1)y^2\} \equiv \{1, 3\} \pmod 4. \tag{4}$$

On the other hand,

$$X^3 + 6 \equiv 1 \pmod 4 \tag{5}$$

for $X \equiv 1 \pmod 4$. Factor the left-hand side as

$$X^3 + 6 = (X+2)(X^2 - 2X + 4). \tag{6}$$

Because $X \equiv 1 \pmod 4$, both factors on the right-hand side of (5) are odd and congruent to 3 modulo 4.

A short gcd computation gives

$$\begin{aligned} \gcd(X+2, X^2 - 2X + 4) &= \gcd(X+2, (X^2 - 2X + 4) - (X-4)(X+2)) \\ &= \gcd(X+2, 12). \end{aligned}$$

Since $X \equiv 1 \pmod 4$, $X + 2 \equiv 3 \pmod 4$. If $3 \mid X + 2$, then $X \equiv 1 \pmod 3$ and therefore $X^3 + 6 \equiv 1 \pmod 3$. But with z even we have $z^2 - 1 \equiv 0$ or $2 \pmod 3$, and with y odd we have $y^2 \equiv 1 \pmod 3$. Hence the right-hand side of (2) is congruent to 0 or 2 modulo 3, a contradiction. Thus $3 \nmid X + 2$, and

$$\gcd(X+2, X^2 - 2X + 4) = 1. \tag{7}$$

Now compare (5) and (2). The right-hand side of (2) factors as

$$y^2(z-1)(z+1),$$

and because $z - 1$ and $z + 1$ are coprime, their prime divisors must be split between the coprime factors $(X+2)$ and (X^2-2X+4). Since y^2 is a square and every odd square is congruent to 1 modulo 4, the two factors on the right-hand side must be congruent to 1 and 3 modulo 4, respectively. By contrast, the two factors on the left-hand side of (5) are both congruent to 3 modulo 4. This is impossible.

Conclusion. The four exhaustive cases all fail. Therefore there are no integers x, y, z with z even, y odd, and $x \equiv 3 \pmod 4$ satisfying the given equation.